DIFFERENTIALS ARE STABLE UNDER ARBITRARY BASE CHANGE FOR ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1203-1217 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1219-1224, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. No quasi-compactness or finiteness hypothesis: the isomorphism holds for every base change.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The construction of the canonical comparison map c_f (lemma-functoriality-differentials, whose own proof is a pointer and is not collected; the construction is a standard special case of functoriality of differentials for ringed topoi), etale-locality of the claim (lemma-localize-differentials), and the schemes case (Morphisms, lemma-base-change-differentials).

Source reference. Original lemma 04CX: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1062–1083. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04CU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 981–996. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01V0: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1224–1224. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $X \to Y$ be a morphism of algebraic spaces over S. Let $g : Y' \to Y$ be a morphism of algebraic spaces over S. Let $X' = X_{Y'}$ be the base change of X. Denote $g' : X' \to X$ the projection. Then the map*

$$(g')^*\Omega_{X/Y} \to \Omega_{X'/Y'}$$

of Lemma 04CX is an isomorphism.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Follows from the schemes version, see Morphisms, Lemma 01V0 and étale localization, see Lemma 04CU. □